Causal Thinking

Unit 9: From “What Goes With What” to “What Happens If We Change It”

Stat 220 | Statistics for Data Science

Where this fits

- Every model so far answers an *association* question: given what I observe, what do I expect?
- Every decision anyone actually needs is a *causal* question: if we change the price, send the email, launch the program, what will happen?
- These are different questions, and no amount of model accuracy converts one into the other.

Principle: prediction and intervention are not the same job

A model can predict rain beautifully from barometer readings. Smashing the barometer does not stop the rain.

Live experiment: manufacture a correlation out of nothing

Live demo: everyone roll two dice

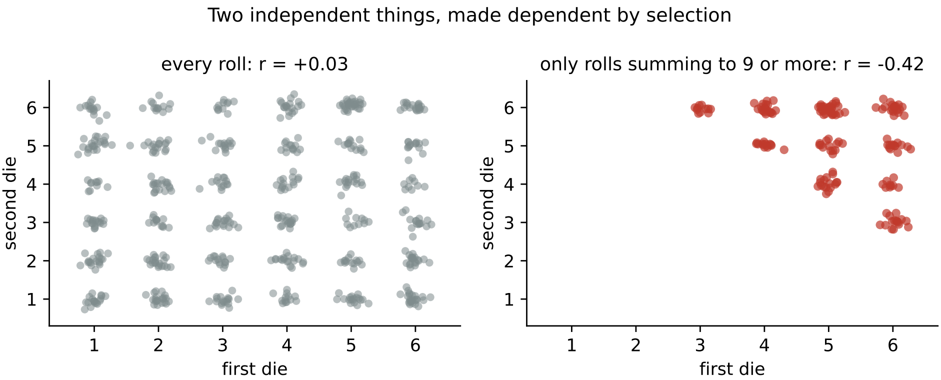
Use real dice, or your phone. Roll **two** dice and report both numbers, in order.

- I will plot every pair on the board. First die on the horizontal axis, second on the vertical.
- The two rolls are obviously independent. Nobody disputes that.

Now: **keep only the pairs whose sum is 9 or more**, and look at the plot again.

Materials: dice, or a phone. Time: 10 minutes. Predict, before we do it: what will the relationship look like in the surviving pairs?

The reveal: selection created a relationship



Independent to begin with. Among high-sum rolls, a low first die *requires* a high second die, so the two become strongly negatively related. Nothing about the dice changed. We changed who was in the dataset.

Where this shows up in real data

- Restaurants you hear about are either great food or great location, because bad on both close. So in your data, food quality and location look negatively related.
- Admitted students are either strong on tests or strong on essays. Within the admitted class, the two look like they trade off.
- Hospitalized patients have some serious condition, so among them, unrelated diseases appear connected.

Principle: the sample is part of the analysis

Before you interpret any relationship, ask who got into the dataset, and what it took to get in.

What this unit gives you

Things to know

- what a causal effect is, in terms of what would have happened otherwise
- why randomization works, and precisely what it buys
- what a confounder is, and why you must control for it
- what a collider is, and why controlling for it is a disaster
- what a mediator is, and when controlling for it answers the wrong question
- the main quasi-experimental designs and the assumption each one leans on

Mistakes to stop making

- reading a regression coefficient as the effect of intervening
- controlling for a collider or for a variable on the causal path
- comparing volunteers to non-volunteers and calling it an effect
- ignoring who selected into the dataset
- treating a before-and-after change as an effect with no control group
- forgetting that the treatment might be caused by the outcome

An effect is a comparison with something that never happened

- For one customer, the effect of the discount is: what they spend *with* it, minus what they would have spent *without* it.
- You can only ever observe one of those two. The other is a counterfactual, and it is missing by definition.
- So causal inference is a **missing data problem**: everything hinges on how you fill in the world that did not happen.

Principle: the question behind every causal claim

"Compared to what?" If someone cannot tell you what their comparison group is, they do not have a causal estimate.

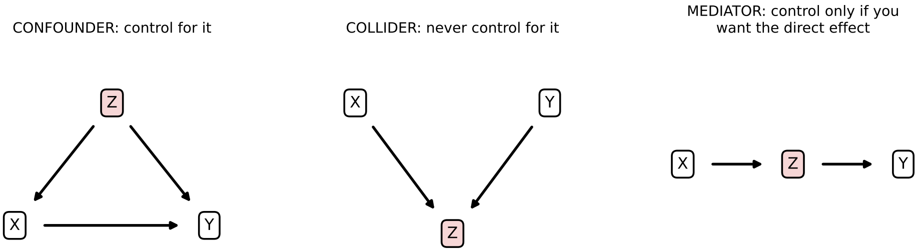
Why randomization is the gold standard

- Randomly assigning treatment makes the two groups comparable *in expectation* on everything: age, motivation, prior spending, and every variable you never thought to measure.
- That last clause is the important one. Adjustment can only handle confounders you measured. Randomization handles the ones you did not.
- What it does *not* fix: small samples, dropout after assignment, non-compliance, contaminated control groups, or measuring the wrong outcome.

In practice: say this precisely

"Randomization does not make the groups identical. It makes any difference between them the kind of difference we can quantify with a standard error."

The three roles a third variable can play



The arrows are what you believe about the world. Which variables to control for is decided by these diagrams, *not* by which ones improve the fit.

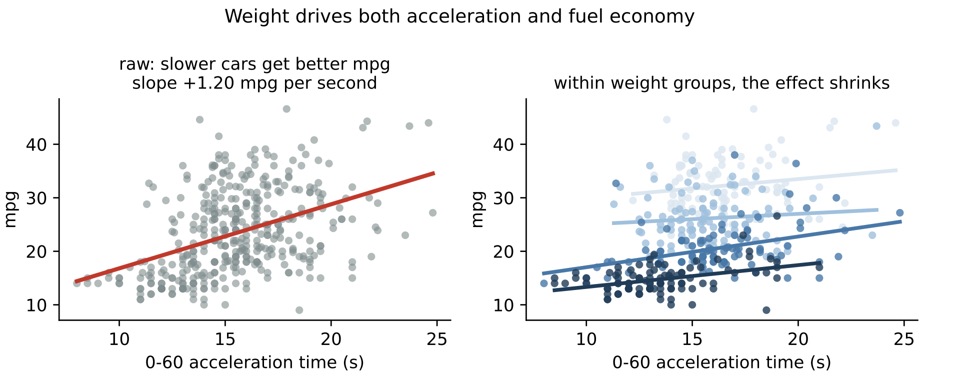
Confounders: the classic problem

- A **confounder** causes both the treatment and the outcome, producing an association with no causal path behind it.
- Ice cream sales and drownings both rise with temperature. Neither causes the other.
- In business: customers who *choose* the premium plan were already more engaged, so premium looks like it causes retention.

Principle: control for confounders, and say which ones you could not measure

The honest sentence is: "adjusting for tenure, plan size, and industry, the difference is X, but I cannot rule out that more motivated teams chose the feature."

Confounding in real data



392 cars. Slower-accelerating cars average **1.20 mpg better per extra second** to 60. Once you adjust for weight, which drives both, the association collapses to **0.26**. Nearly 80% of the raw relationship was weight, relabeled.

Colliders: the problem nobody warns you about

- A **collider** is caused by both variables. Conditioning on it creates an association where none existed, exactly as the dice demonstrated.
- Selecting your sample on a collider is the same mistake: analyzing only customers who *contacted support*, only patients who were *admitted*, only startups that *got funded*.
- This is why “add every variable you have to the regression” is bad advice. Some of those variables are colliders, and adding them injects bias rather than removing it.

Trap: the kitchen-sink regression

More controls is not safer. Each control is a claim about the causal diagram, and the wrong one makes your estimate worse in a direction you cannot sign.

Mediators: controlling away the thing you wanted to measure

- A **mediator** sits on the causal path: training raises skill, skill raises sales.
- If you regress sales on training *controlling for skill*, you have removed the very channel through which training works, and the effect vanishes.
- Controlling for a mediator answers a different, narrower question: is there a direct effect *other than* through skill?

In practice: a question that trips people up

"Should you control for everything?" The right answer is no, and the reason is exactly this: confounders yes, colliders never, mediators only if you deliberately want the direct effect.

The rules, in one table

Role	Structure	What to do
Confounder	causes both X and Y	**Control for it.** Omitting it biases the estimate.
Collider	is caused by both X and Y	**Never control for it**, and never select the sample on it.
Mediator	X causes it, it causes Y	**Leave it out** for the total effect.

Principle: the order of operations

Decide the role from what you believe about the world, *then* write the model. You cannot read the role off a correlation matrix, and the fit will not tell you either.

Your turn #1

Your turn: classify the variable

You want the effect of a **new onboarding flow** on 90-day retention. For each variable, say whether you should control for it, and why:

1. Company size at signup.
2. Number of features used in week one.
3. Whether the customer contacted support at any point.

Your turn #1: answer

- **Company size:** a confounder. It affects both which flow they got (if assignment was not random) and retention. **Control for it.**
- **Features used in week one:** a mediator. That is how onboarding works. **Do not control** unless you specifically want the effect that runs through some other channel.
- **Contacted support:** a collider. It is caused by both the onboarding experience and by the customer's underlying trouble. **Never control for it**, and do not restrict the sample to those who contacted support.

Principle: draw the arrows before you write the formula

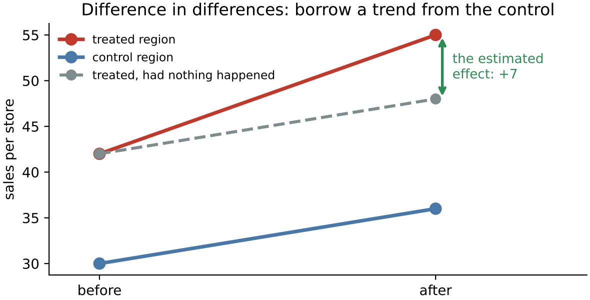
Two minutes with a diagram prevents a month of arguing about a coefficient.

The realistic situation

Often you cannot randomize: it is unethical, illegal, too slow, or the change already happened. Four workhorse designs, each buying credibility with an assumption:

- **Matching / adjustment:** compare treated units to similar untreated ones. *Assumes* you measured every confounder.
- **Difference in differences:** compare changes over time against a control group. *Assumes* the two groups would have moved in parallel.
- **Regression discontinuity:** compare units just above and just below a cutoff. *Assumes* units cannot precisely manipulate their position.
- **Instrumental variables:** use a nudge that affects treatment but nothing else. *Assumes* the instrument has no other path to the outcome.

Difference in differences, drawn



The treated region rose by 13. But the control region rose by 6 without any treatment, so the economy was moving anyway.

- Estimated effect: $13 - 6 =$ **7**.
- Everything rests on the dashed line, the claim that the treated region would have followed the control's trend.

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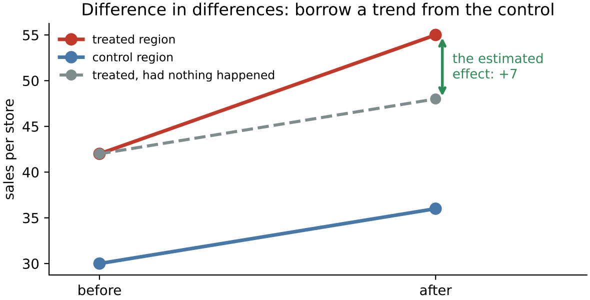

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So the first thing to check is whether the two lines were parallel *before* the change.

Difference in differences, with the actual arithmetic

A chain raises pay in its **Ohio** stores in July and leaves **Indiana** alone.

	before	**after**	**change**
Ohio (treated)	42.0	55.0	$+13.0$
Indiana (control)	30.0	36.0	$+6.0$
difference in differences			$\mathbf{+7.0}$

- Ohio rose 13, but 6 of that would have happened anyway. The estimated effect is **7**.
- It all rests on the claim that Ohio *would have* risen by 6 too, which is untestable after treatment.

Principle: how to check what you cannot test

Plot both series for several periods *before* the change. If they moved together then, the assumption is credible.

Reverse causation, and how to rule it out

- Hospitals with more staff have higher mortality. Do nurses kill people, or do sicker patients get more nurses?
- Ad spend and sales move together, partly because budgets are *set* as a percentage of expected sales.
- Police numbers and crime, tutoring and grades, maintenance and failures: in each case the outcome plausibly causes the treatment.

Principle: the fix is timing, not statistics

Establish that the cause moved *first*, and for a reason unrelated to the outcome. That is what a rollout date, a policy change, or a budget freeze gives you, and no regression can substitute for it.

Natural experiments worth knowing

- **John Snow, 1854:** two water companies served interleaved houses on the same London streets, one drawing from sewage-contaminated water. The houses were otherwise alike, so the cholera difference was as good as randomized.
- **Minimum wage, 1994:** Card and Krueger compared fast-food employment in New Jersey after a wage increase against neighboring Pennsylvania, a difference in differences that reversed the textbook prediction.
- **Scholarship cutoffs:** students just above and just below a score threshold are nearly identical, so the cutoff acts like a coin flip.

Principle: look for the accident

The best observational studies find a moment where something outside the actors' control assigned the treatment for them.

Your turn #2

Your turn: design the study

Your company rolled out a new sales-training program. Reps who completed it sold 18% more last quarter than reps who did not. Leadership wants to make it mandatory.

With a neighbor: name two reasons that 18% may not be the effect of training, and design a study you could actually run next quarter.

Your turn #2: answer

- **Selection:** reps who volunteered were more motivated, and would have improved anyway. **Regression to the mean:** if struggling reps were assigned, expect a bounce with no cause.
- **Timing:** the quarter may have been better for everyone, which a simple before-after comparison would credit to training.
- **The study:** randomize the next cohort. If that is impossible, use a staged rollout by region and compare with difference in differences, checking that the regions were parallel beforehand.

Principle: staged rollouts are free experiments

If you must launch to everyone eventually, launch in a random order. It costs nothing and converts a guess into an estimate.

Your turn #3

Your turn: sort the variables

You want the effect of a **price increase** on **customer churn**. For each variable, say confounder, collider, or mediator, and whether it belongs in the model:

1. The customer's contract tier, which affects both the price they were offered and their likelihood of leaving.
2. Whether the customer called support to complain about the price.
3. How much the customer's monthly bill actually rose.

Your turn #3: answer

- **Contract tier: confounder.** It drives both the treatment and the outcome, so leaving it out biases the estimate. **Control for it.**
- **Complaint call: collider.** It is caused by the price increase *and* by the customer's underlying unhappiness. Controlling for it, or studying only complainers, manufactures a relationship. **Leave it out.**
- **Size of the bill increase: mediator.** It is how the price increase acts. Include it only if you specifically want the effect of price that runs through something else.

Principle: how to spot it

Ask "did this happen before or after the treatment?" Anything measured afterward is a candidate mediator or collider, and both are dangerous controls.

LLM check: mistake or not? (1 of 2)

LLM check: we asked an LLM about a feature launch

We told it: "Customers who use our new dashboard renew at 91%, others at 68%. What is the effect of the dashboard on renewal?"

It answered: "The dashboard increases renewal by 23 percentage points. To be thorough I would control for every variable available in your database to remove any confounding."

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Mistake, or no mistake?

Trap: verdict: mistake, twice

First, the 23 points is a comparison between customers who *chose* the dashboard and those who did not, so it is mostly selection. Second, "control for everything" is wrong: some of those variables are colliders and mediators, and controlling for them adds bias.

LLM check: mistake or not? (2 of 2)

LLM check: same territory, a different answer

We told it: "Our ad spend and sales are correlated at 0.8 across 24 months. Does advertising drive sales?"

It answered: "Not necessarily. Ad budgets are often set as a percentage of expected sales, so sales may be causing spend. Both also rise in the holiday season, which is a confounder. I would look for a period when spend changed for a reason unrelated to sales, such as a budget freeze or a regional rollout, and compare against a similar region."

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Mistake, or no mistake?

Principle: verdict: no mistake

It raised reverse causation, named a confounder, and proposed a design. That is the full causal checklist in three sentences.

How to answer "does X cause Y?"

1. **State the intervention precisely.** Who gets what, instead of what?
2. **Name the comparison group.** If there is none, stop.
3. **Draw the diagram**: candidate confounders, colliders, mediators.
4. **Ask how treatment was assigned.** Randomized, chosen, or assigned by a rule?
5. **Choose the design** that fits the assignment mechanism.
6. **Check the design's assumption** explicitly (parallel pre-trends, balance, no manipulation at the cutoff).
7. **Report the effect with its uncertainty and its caveats**, including the confounder you could not measure.

Scenario: the loyalty program that "worked"

Setup. Members of a new loyalty program spend \$340 a year. Non-members spend \$180. Marketing claims the program doubled spending.

- Who joins a loyalty program? People who already shop there often. The comparison is between heavy and light customers, not between two versions of the same customer.
- Adjusting for prior-year spending would help, but only if you have it, and it will not remove intent.
- The clean answer: randomize invitations, or use a staged geographic rollout, and compare the change in spending, not the level.

You may be asked: how to answer

"The 160-dollar gap is mostly selection. I would estimate this from a randomized invitation, and I would look at the change in spending rather than its level."

Answering “does X cause Y?”

1. **State the intervention precisely.** Who gets what, instead of what?
2. **Name the comparison group.** If there is none, stop.
3. **Draw the diagram** and classify every candidate control as confounder, collider, or mediator.
4. **Ask how treatment was assigned**: randomized, self-selected, or by a rule?
5. **Choose the design that matches the assignment**, and write down the assumption it buys.
6. **Check that assumption explicitly**, for example by plotting pre-treatment trends.
7. **Report the estimate, its uncertainty, and the confounder you could not measure.**

The traps, all in one place

Trap: the ones that bite most often

- Reading a regression coefficient as the effect of intervening.
- Controlling for everything you have.
- Conditioning on a collider, including by how you selected the sample.
- Controlling for a mediator and reporting the shrunken effect as "no effect."
- Comparing volunteers to non-volunteers.
- Treating a before-after change as an effect with no control group.
- Forgetting that the treatment might be caused by the outcome.

Ten questions to be able to answer

1. Users of feature X retain better. Should we push everyone onto it?
2. Why does randomization work, and what does it fail to fix?
3. Should you control for every variable you have? Explain with an example.
4. What is a collider, and give a case where selecting on one misled someone?
5. When would controlling for a variable be actively harmful?
6. We cannot run an experiment. What are the options, and what does each assume?
7. How would you check the parallel-trends assumption?
8. Ad spend and sales correlate at 0.8. What do you conclude?
9. Our program's participants improved 18%. What do you ask first?
10. How would you design a rollout so it produces a causal estimate for free?

Mastery checklist

You have this unit when you can, from memory:

- define a causal effect as a comparison with a counterfactual
- explain precisely what randomization buys and what it does not
- classify a variable as confounder, collider, or mediator and act accordingly
- explain why “control for everything” is wrong
- name four quasi-experimental designs and the assumption behind each
- design a staged rollout that yields a defensible estimate

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	**What you would reach for**
Adjusted comparison	`smf.ols(ÿ ˜ treat + covariates; data)`
Estimates within strata	`pd.qcut` then `groupby`
Difference in differences	four group means, then the difference of the differences
Check parallel pre-trends	plot both series for several periods before treatment
Randomized assignment for a rollout	`rng.integers(0, 2, n)` on the unit of assignment
Simulate a confounder or a collider	generate the data yourself and compare estimates
Logistic version for a binary outcome	`smf.logit`

If you cannot say what the output means, the fact that you produced it is worth nothing.

Where we go next

- We have been treating the data as the only source of information. **Unit 10** lets prior knowledge enter the estimate explicitly, and makes decisions by weighing outcomes rather than by rejecting a null.

Principle: the through-line

The data can only tell you what went with what. The causal claim comes from the design, and the design is your job.